

Test Specifications: Events Domain

Gema Event Management Platform
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Domain: Events

Source files:

- backend/src/routes/event.routes.ts
- backend/src/models/Event.ts
- backend/src/services/event.service.ts

Scope

Covers the full lifecycle of the `Event` resource: public discovery, vendor-owned CRUD, admin moderation, capacity/availability logic, status transitions, ownership enforcement, search/filtering, and unauthenticated access to protected routes.

Test Cases

Test ID	Description	Preconditions	Input	Expected Output	Priority
TC-EVT-001	List published events (public)	At least one published, approved, active event in DB	GET /api/events	200 OK; array of published events; pagination metadata present	High
TC-EVT-002	Get event by valid slug (public)	Event with slug <code>summer-olympiad-2026</code> exists and is published	GET /api/events/summer-olympiad-2026	200 OK; full event document returned	High
TC-EVT-003	Get event by non-existent slug	No event with given slug	GET /api/events/does-not-exist	404 Not Found	High
TC-EVT-004	Get event categories (public)	Categories exist in DB	GET /api/events/categories	200 OK; array of category strings	Medium
TC-EVT-005	Get unique cities (public)	Events with varied cities exist	GET /api/events/cities	200 OK; deduplicated array of city strings	Medium
TC-EVT-006	Create event as authenticated vendor	Vendor JWT in cookie; valid body	POST /api/events with all required fields (title, description, category, type, venueType, ageRange, location, price, currency, dateSchedule)	201 Created; event document with status: "draft", vendorId set to requesting vendor	High
TC-EVT-007	Create event — missing required field (title)	Vendor JWT in cookie	POST /api/events body omitting title	400 Bad Request; validation error referencing title	High
TC-EVT-008	Create event — invalid type enum value	Vendor JWT in cookie	POST /api/events with type: "Hackathon"	400 Bad Request; validation error listing accepted type values	Medium
TC-EVT-009	Create event — schedule date in the past	Vendor JWT in cookie; otherwise valid body	dateSchedule[0].startDate set to yesterday	400 Bad Request; error "Schedule date cannot be in the past"	High
TC-EVT-010	Create event — endDate before startDate	Vendor JWT in cookie	dateSchedule[0].startDate = today+2, endDate = today+1	400 Bad Request; error "End date must be after start date"	Medium
TC-EVT-011	Create event — ageRange invalid (min > max)	Vendor JWT in cookie	ageRange: [18, 10]	400 Bad Request; error "Invalid age range values"	Medium
TC-EVT-012	Create event — tags exceeds 20-item limit	Vendor JWT in cookie	tags array with 21 items	400 Bad Request; error "Cannot have more than 20 tags"	Low
TC-EVT-	Create event — images exceeds 10-	Vendor JWT in cookie	images array with 11 items	400 Bad Request; error "Cannot have more than 10	Low

013	item limit				images"	
TC- EVT- 014	Create event — unauthenticated request	No auth cookie / header	POST /api/events with valid body	401 Unauthorized	High	
TC- EVT- 015	Create event — authenticated as non-vendor role (e.g., customer)	Customer JWT in cookie	POST /api/events with valid body	403 Forbidden	High	
TC- EVT- 016	Update own event as vendor	Vendor JWT; event owned by that vendor	PUT /api/events/:id with { "title": "Updated Title" }	200 OK; updated event document; title reflects new value	High	
TC- EVT- 017	Update another vendor's event	Vendor JWT; event owned by a different vendor	PUT /api/events/:id targeting another vendor's event	403 Forbidden	High	
TC- EVT- 018	Update event — admin can update any event	Admin JWT	PUT /api/events/:id targeting any vendor's event	200 OK; event updated regardless of vendorId	High	
TC- EVT- 019	Update event — invalid Mongo ID format	Vendor JWT	PUT /api/events/not-a-mongo-id	400 Bad Request; validation error "Invalid event ID"	Medium	
TC- EVT- 020	Update event — set price to negative value	Vendor JWT; event owned by vendor	PUT /api/events/:id with { "price": -5 }	400 Bad Request; error "Price must be a positive number"	Medium	
TC- EVT- 021	Delete own event as vendor	Vendor JWT; event owned by that vendor	DELETE /api/events/:id	200 OK; event soft-deleted (isDeleted: true, deletedAt set); no longer returned in public listing	High	
TC- EVT- 022	Delete another vendor's event	Vendor JWT; event owned by different vendor	DELETE /api/events/:id	403 Forbidden	High	
TC- EVT- 023	Capacity check — event sold out	All dateSchedule entries have availableSeats: 0 and unlimitedSeats: false	GET /api/events/:slug	200 OK; response indicates sold-out status; booking attempt returns 400/409	High	
TC- EVT- 024	Capacity check — unlimited seats flag	Schedule entry has unlimitedSeats: true	POST /api/events with unlimitedSeats: true, no availableSeats field	201 Created; no seat-limit validation error triggered	Medium	
TC- EVT- 025	Status transition — publish a draft event	Admin JWT; event in status: "draft"	PUT /admin/:id/approval with { "isApproved": true }	200 OK; event isApproved: true; status transitions to published	High	
TC- EVT- 026	Status transition — reject an event with reason	Admin JWT; event pending approval	PUT /admin/:id/approval with { "isApproved": false, "reason": "Content policy violation" }	200 OK; event isApproved: false, status set to rejected, reason stored	High	
TC- EVT- 027	Admin approval — reason exceeds 500 chars	Admin JWT	PUT /admin/:id/approval with reason of 501 characters	400 Bad Request; validation error on reason field	Low	
TC- EVT- 028	Admin: get all events (including drafts/rejected)	Admin JWT	GET /api/events/admin/all	200 OK; returns events of all statuses including draft, pending, rejected, archived	High	
TC- EVT- 029	Admin: get all events — non- admin role	Vendor JWT	GET /api/events/admin/all	403 Forbidden	High	
TC- EVT- 030	Toggle event featured status	Admin JWT; event exists	PUT /api/events/admin/:id/featured	200 OK; isFeatured toggled (true -> false or false -> true)	Medium	
TC- EVT- 031	Vendor: list own events	Vendor JWT	GET /api/events/vendor/my-events	200 OK; only events where vendorId matches the requesting vendor	High	
TC- EVT- 032	Vendor analytics — unauthenticated	No auth	GET /api/events/vendor/analytics	401 Unauthorized	High	
TC- EVT- 033	Claim event as vendor	Vendor JWT; unclaimed event exists	POST /api/events/:id/claim	200 OK; event vendorId set to requesting vendor	Medium	
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EVT-Claim event — non-Customer JWT 034	vendor role		POST /api/events/:id/claim	403 Forbidden	Medium
TC-EVT-035	Search/filter events by city	Published events in multiple cities	GET /api/events?city=Dubai	200 OK; results contain only events where location.city = "Dubai"	High
TC-EVT-036	Search/filter events by category	Published events in multiple categories	GET /api/events?category=Sports	200 OK; all returned events have category = "Sports"	High
TC-EVT-037	Search/filter events — expired events excluded	Events with all dateSchedule dates in the past	GET /api/events (public listing)	200 OK; expired events not included in default listing	High
TC-EVT-038	Create event with invalid currency	Vendor JWT	POST /api/events with currency: "GBP"	400 Bad Request; error listing accepted currencies (AED, EGP, CAD, USD)	Medium
TC-EVT-039	Create event with invalid coordinates	Vendor JWT	POST /api/events with location.coordinates.lat: 200 (out of range)	400 Bad Request; error "Invalid latitude"	Low
TC-EVT-040	MediaAsset validation — non-existent asset ID	Vendor JWT	POST /api/events with imageAssets: ["<nonExistentObjectId>"]	400/404; error "MediaAsset not found"; transaction rolled back	Medium

Coverage Summary

Area	Test IDs	Count
Public read (list, get, categories, cities)	TC-EVT-001 to TC-EVT-005	5
Create — happy path	TC-EVT-006	1
Create — validation errors	TC-EVT-007 to TC-EVT-013	7
Create — security (unauth / wrong role)	TC-EVT-014 to TC-EVT-015	2
Update — happy path + ownership	TC-EVT-016 to TC-EVT-019	4
Update — validation errors	TC-EVT-020	1
Delete — ownership + security	TC-EVT-021 to TC-EVT-022	2
Capacity / availability	TC-EVT-023 to TC-EVT-024	2
Status transitions (approval flow)	TC-EVT-025 to TC-EVT-027	3
Admin moderation + feature toggle	TC-EVT-028 to TC-EVT-030	3
Vendor-specific endpoints	TC-EVT-031 to TC-EVT-034	4
Search and filtering	TC-EVT-035 to TC-EVT-037	3
Miscellaneous edge cases	TC-EVT-038 to TC-EVT-040	3
Total		40

Notes

- "Soft delete" behaviour (isDeleted, deletedAt) is inferred from the Event model. Tests assume the public listing query filters isDeleted: false.
- status field has five values: draft, published, archived, pending, rejected. TC-EVT-025/026 cover the admin approval path; transitions to archived are not currently exposed via a documented route and should be added when that endpoint ships.
- The claimEvent endpoint (TC-EVT-033) semantics (ownership transfer vs. initial claim) should be confirmed against controller logic before finalising acceptance criteria.
- TC-EVT-023 assumes a downstream booking service enforces the sold-out check; if that guard lives in the event service itself, update the expected HTTP status accordingly.
- All authenticated test cases should be run twice: once with a valid JWT cookie and once with a valid Authorization: Bearer header, to verify the dual-auth fallback.